Soap bubbles, then deeper

Soap Bubbles

Name _____ Date _____



Little Wonderer dips a wand and blows once. A round soap bubble lifts into the soft garden light. Big Wonderer watches beside the little one. The bubble looks simple, yet its wall is busy. What keeps that thin shine from falling apart right away?



A soap bubble is a thin film of water mixed with soap, wrapped around air. Soap molecules help the watery film hold together longer than plain water would. You can stretch a flat film on a wand before any bubble forms. The wall looks delicate because it truly is thin.



Light reflects from the front and back of the soap film. Those reflections can add or cancel, so stripes of color appear. As water drains or evaporates, the film thins and the colors shift. Scientists call this thin-film interference. The swirl is a clue, not paint on the bubble.

soap film	a thin watery wall mixed with soap
interference	light waves adding or canceling after reflecting
surface tension	the pull along a liquid's surface
evaporate	when liquid water turns into vapor in the air
sphere	a round ball shape with even sides

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Big Wonderer sketches the moving color stripes while the little one watches. A bright belt may slide as the film changes thickness. Recording what you see turns a pretty swirl into evidence. Later you can ask whether the top of the bubble thinned faster than the bottom.



They look again at a free bubble beside a stretched film on a wand. Surface tension pulls on the soap film from every side. For the air inside, a sphere uses the least film area, so a free bubble settles into a round shape. The wand can force other shapes until the air is trapped.



Every ordinary soap bubble has a short life. Touch, dust, or a dry thin spot can open a hole in the film. Even in still air, water leaves the wall as it drains and evaporates. When the film breaks, tiny droplets scatter and the sphere is gone. What might help a bubble last a little longer?

1. What do you notice first about a floating soap bubble?
2. What is the thin wall of a soap bubble made of?
3. Why can colors swirl and shift on a bubble?
4. Why are free-floating bubbles often round, and why do they pop?

Your turn

Write or dictate three sentences: what a bubble wall is made of, why colors can swirl, and one careful way to watch a bubble without poking it toward a face.

Example: The wall is a thin soap film around air. Colors shift as the film thins. I will watch from the side and keep wands away from eyes.